



TRINITY GRAMMAR SCHOOL
MATHEMATICS DEPARTMENT



MATHEMATICS

(2 Unit / Extension 1)

YEAR 11 2005 YEARLY EXAMINATION

Assessment Task 4

Weighting: 40%

Time Allowed: *two and a half hours*
(plus 5 minutes reading time)

Outcomes referred to: P1, P2, P3, P4, P5, P6, P7, P8

Question	Topic	Mark	Max. Mark
1	Basic Arithmetic and the Calculator		10
2	Basic Algebra and Surds		10
3	Factorisation and Equations		10
4	Geometry		10
5	Functions		10
6	Trigonometry		10
7	Trigonometry		10
8	The Linear Function		10
9	Calculus		10
10	Calculus		10

Instructions:

1. Board Approved Calculators may be used.
2. Show all working.
3. Each question is 10 marks. Mark values are shown beside each part.
4. Each question must be answered on a separate page. Write your name and teacher on the top of each page.

QUESTION 1 (10 Marks)

- a) Evaluate $\frac{45.8 \times 7.6}{22.9}$ (1 mark)
- b) Round off 0.0276 (to 2 significant figures) (1 mark)
- c) Write $(3.1 \times 10^5)^2$ in scientific notation (1 mark)
- d) Write $\frac{91}{208}$ as a percentage correct to 1 d.p. (1 mark)
- e) Find 110% of \$63.74 (1 mark)
- f) What percentage is 35c of \$1.12? (2 marks)
- g) Express $\frac{8}{45}$ as a recurring decimal. (1 mark)
- h) Express $0.3\bar{3}\bar{7}$ as a common fraction. (2 marks)

QUESTION 2 (10 Marks) *Begin a new page*

- a) Simplify $\sqrt[3]{125}$ (1 mark)
- b) Simplify $(2\sqrt{7})^3$ (2 marks)
- c) Write $(1 - 2\sqrt{7})^2$ in the form of $a + b\sqrt{x}$ (2 marks)
- d) Rationalise the denominator of $\frac{\sqrt{7} + 1}{\sqrt{7}}$ (2 marks)
- e) Simplify $x^2y^{-3} \times x^{-4}y^9$ (1 mark)
- f) Write $5\sqrt{5}$ in the form 5^n (1 mark)
- g) Expand and simplify $(8r - 5s)(8r + 5s)$ (1 mark)

QUESTION 3 (10 Marks) *Begin a new page*

a) Factorise

i) $7n - 14nm$ (1 mark)

ii) $3x^2 + 11x + 6$ (1 mark)

iii) $a^3 + 8$ (1 mark)

iv) $25x^2 - 9$ (1 mark)

b) Simplify

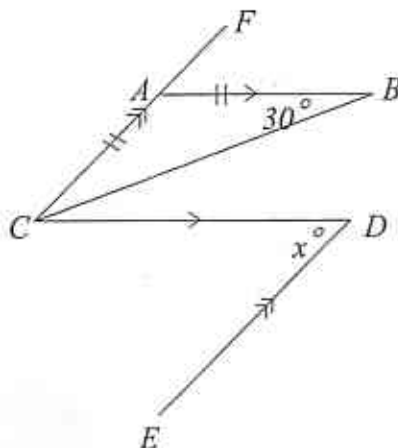
i) $\frac{3x^2 + 13x - 10}{x^2 + 5x}$ (2 marks)

ii) $\frac{a^2b}{cd} \div \frac{ab^2}{c^2d}$ (2 marks)

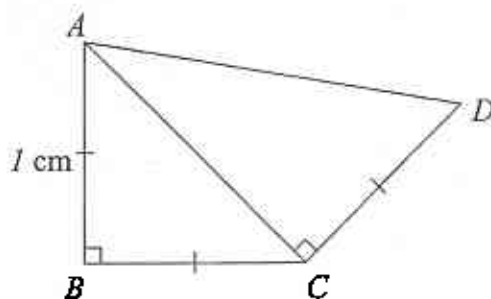
c) Solve $2x^2 - 3x + 1 = 0$ (2 marks)

QUESTION 4 (10 Marks) *Begin a new page*

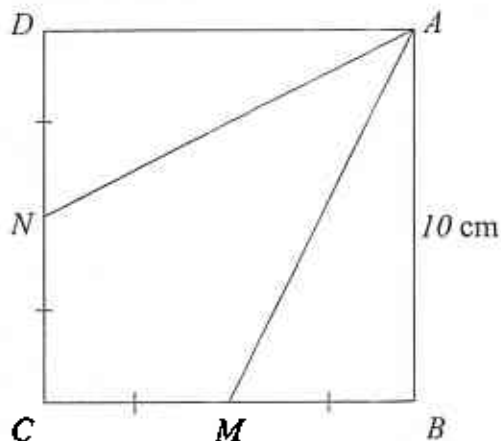
- a) In the diagram given below, $AB \parallel CD$, $CA \parallel ED$ and $AB = AC$. Find the value of x , giving reasons. (3 marks)



- b) Prove that $AD = \sqrt{3}$ cm. (2 marks)



- c) $ABCD$ is a square of side length 10 cm. M is the midpoint of BC and N is the midpoint of CD .



- Prove that $\triangle ABM = \triangle ADN$. (3 marks)
- What type of quadrilateral is the figure $AMCN$? Give reasons. (2 marks)

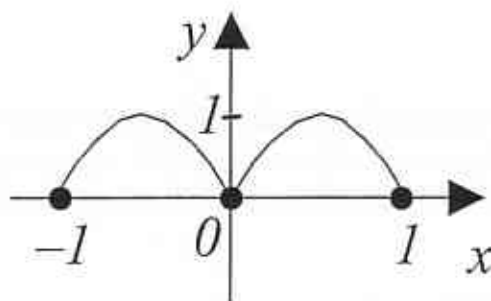
QUESTION 5 (10 Marks) *Begin a new page*

a) For the function $f(x) = 2x - 1$, find:

i) the value of $f(2)$ (1 mark)

ii) the value of x at which $f(x) = 0$ (1 mark)

b) Find the domain (D) and the range (R) of the function below. (2 marks)



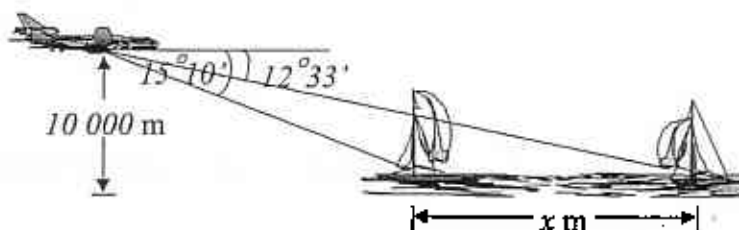
c) State whether the function $y = \frac{1}{4-x} + \frac{1}{4+x}$ is even, odd or neither. Show working. (2 marks)

d) Sketch the function $y = 1 - x^2$ in the domain $-1 \leq x \leq 1$ (2 marks)

e) Shade the region(s) in the XY -plane indicated by the inequality $x^2 + y^2 < 1$ (2 marks)

QUESTION 6 (10 Marks) *Begin a new page*

- a) In which quadrants does θ lie if $\cos \theta$ is negative and $\tan \theta$ is negative? (1 mark)
- b) Find the exact value of $\frac{\cos 45^\circ}{\tan 30^\circ}$ (2 marks)
- c) Prove that $\frac{\cot \theta \cos \theta}{1 + \cot^2 \theta} = \cos \theta$ (3 marks)
- d) From a plane flying at a constant altitude of 10 000 m, the angle of depression of two boats is observed. The angle of depression of boat A is $15^\circ 10'$ and at the same instant the angle of depression of boat B is $12^\circ 33'$. Find (to the nearest metre) the distance x between the two boats. (2 marks)

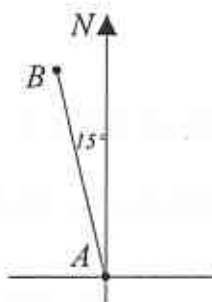


- e) Find the exact value of $\tan(-150)^\circ$ (2 marks)

QUESTION 7 (10 Marks) *Begin a new page*

- a) Write down the bearing of B from A .

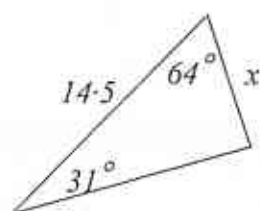
(1 mark)



- b) Find, to the nearest degree, all possible values of θ if $\sin \theta = 0.6018$ and $0 \leq \theta \leq 360^\circ$ (2 marks)

- c) Find x (to 1 decimal place).

(2 marks)



- d) In $\triangle ABC$, $\angle A = 36^\circ$, $a = 11.7$ and $b = 9.6$. Find $\angle B$ (to the nearest degree). (2 marks)

- e) A yacht sails due south of its home port for 175 nautical miles. It then turns and sails for 350 nautical miles on a course bearing 210° . How far (in a straight line) is the yacht from its home port? Give your answer to the nearest nautical mile. (3 marks)

QUESTION 8 (10 Marks) *Begin a new page*

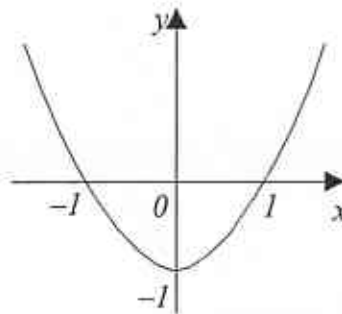
- a) Sketch the graph of $y = -\frac{1}{2}$ (1 mark)
- b) The points A and B have coordinates $(-3, 2)$ and $(4, 7)$, respectively. Find:
- i) the gradient of the straight line passing through A and B (1 mark)
- ii) the distance AB (correct to 1 decimal place) (1 mark)
- c) Show that the lines $y = 3x + 1$ and $9x - 3y + 2 = 0$ are parallel. (2 marks)
- d) Find the equation of the perpendicular bisector of the interval joining $(2, -5)$ and $(-4, 7)$. Express your answer in general form. (3 marks)
- e) The straight line $3x - 4y + p = 0$ is 2 units from the point $(2, 5)$. Find the two possible values of p . (2 marks)

QUESTION 9 (10 Marks) *Begin a new page*

a) Find $\lim_{x \rightarrow -4} \frac{x^4 - 16}{x^2 - 4}$ (2 marks)

b) Using $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$, find $f'(x)$ if $f(x) = 7x + 2$ (2 marks)

c) Sketch the gradient function $f'(x)$ of the curve below and give its equation. (2 marks)



d) If $f(x) = 12x^3$, find the value of $f'(2)$ (2 marks)

e) Find the gradient of the curve $y = 2x^2 + 3x - 1$ when $x = -2$ (2 marks)



QUESTION 10 (10 Marks) Begin a new page

- a) Differentiate $12 + 7x^5$ with respect to x . (1 mark)
- b) If $y = (2x - 5)^3$, find $\frac{dy}{dx}$. (2 marks)
- c) Find the equations of the tangents to the curve $y = x(x - 1)$ at the points where the curve cuts the x axis. (4 marks)
- d) Find the gradient of the normal to $y = x^2$ at the point $(4, 16)$ (1 mark)
- e) Differentiate $y = \frac{x}{x+1}$ with respect to x . (2 marks)

Question 1

a) $\frac{45.8 \times 7.6}{22.9} = 15.2 \checkmark$

b) $0.0276 = 0.028 \text{ (2sf)} \checkmark$

c) $(3.1 \times 10^5)^2 = 9.61 \times 10^{10} \checkmark$

d) $\frac{91}{208} \times 100\% = 43.75\%$
 $= 43.8\% \text{ (1dp)} \checkmark$

e) $\frac{110}{100} \times \$63.74 = \70.114
 $= \$70.11 \text{ (2dp)} \checkmark$

f) $\frac{350}{1120} \times 100\% = 31.25\% \checkmark$

g) $\frac{8}{45} = 0.17777\ldots$
 $= 0.1\bar{7} \checkmark$

h) let $x = 0.3\bar{3}\bar{7}$
 $10x = 3.\bar{3}\bar{7} \checkmark$
 $1000x = 337.\bar{3}\bar{7}$

$1000x = 10x + 334$

$990x = 334$

$x = \frac{334}{990}$

$= \frac{167}{495} \checkmark$

(10)

2005

Question 2

a) $\sqrt[3]{125} = 5 \checkmark$

b) $(2\sqrt{7})^3 = 2\sqrt{7} \times 2\sqrt{7} \times 2\sqrt{7}$
 $= 8 \times 7 \times \sqrt{7}$
 $= 56\sqrt{7} \checkmark$

c) $(1-2\sqrt{7})^2 = (1-2\sqrt{7})(1-2\sqrt{7}) \checkmark$
 $= 1 - 2\sqrt{7} - 2\sqrt{7} + 28$
 $= 29 - 4\sqrt{7} \checkmark$

d) $\frac{\sqrt{7}+1}{\sqrt{7}} \times \frac{\sqrt{7}}{\sqrt{7}} = \frac{7+\sqrt{7}}{7} \checkmark$

e) $x^2y^{-3} \times x^{-4}y^9 = x^{-2}y^6 \checkmark$

f) $5\sqrt{5} = 5 \times 5^{\frac{1}{2}}$
 $= 5^{\frac{3}{2}} \checkmark$

g) $(8r-5s)(8r+5s) = 64r^2 + 40rs - 40rs - 25s^2$
 $= 64r^2 - 25s^2 \checkmark$

(10)

Question ③

- a) i) $7n - 14nm = 7n(1 - 2m)$ ✓
 ii) $3x^2 + 11x + 6 = (3x + 2)(x + 3)$ ✓
 iii) $a^3 + 8 = (a + 2)(a^2 - 2a + 4)$ ✓
 iv) $25x^2 - 9 = (5x - 3)(5x + 3)$ ✓

b) i) $\frac{3x^2 + 13x - 10}{x^2 + 5x} = \frac{(3x - 2)(x + 5)}{x(x + 5)}$ ✓
 $= \frac{3x - 2}{x}$ ✓

ii) $\frac{a^2b}{cd} \div \frac{ab^2}{c^2d} = \frac{a^2b}{cd} \times \frac{c^2d}{ab^2}$ ✓
 $= \frac{ac}{b}$ ✓

c) $2x^2 - 3x + 1 = 0$

$x = \frac{3 \pm \sqrt{9 - 4(2)(1)}}{2(2)}$
 $= \frac{3 \pm \sqrt{1}}{4}$
 $= \frac{3 \pm 1}{4}$
 $= \frac{4}{4}, \frac{-2}{4}$
 $= 1, -\frac{1}{2}$ ✓
 ✓

10

Question ④

- a) $\angle ACB = 30^\circ$ (base $\angle$ s of isos. $\triangle ABC$ are equal)
 $\angle BCD = 30^\circ$ (alt $\angle$ s, $AB \parallel CD$) ✓
 $\angle CDE = x^\circ = 60^\circ$ (corresponding $\angle$ s $CA \parallel ED$) ✓

b) $(AC)^2 = 1^2 + 1^2$ (pythag.)
 $(AC)^2 = 2$ ✓
 $AC = \sqrt{2}$
 $(AD)^2 = (AC)^2 + (DC)^2$
 $= (\sqrt{2})^2 + (1)^2$
 $= 2 + 1$
 $(AD)^2 = 3$ ✓
 $AD = \sqrt{3}$ as required.

- c) i) In $\triangle$ s ABM and ADN

- i) $AB = AD = 10\text{cm}$ (sides of a square are equal)
 ii) $BM = DN = 5\text{cm}$ (N is the midpt of DC ,
 M is the midpt of CB)
 iii) $\angle ADN = \angle ABM = 90^\circ$ ($\angle$ s in a square)
 $\therefore \triangle ABM \cong \triangle ADN$ (SAS) ✓

- ii) $AMCN$ is a kite ✓ as adjacent sides are equal.
 $AN = AM$ (matching sides on congr. $\triangle$ s)
 $NC = CM$ (N and M are midpts of DC and BC)

10

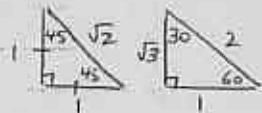
Question 6

a) $\cos \theta < 0$ and $\tan \theta < 0$ in quadrant 2. ✓

b) $\frac{\cos 45^\circ}{\tan 30^\circ} = \frac{\frac{1}{\sqrt{2}}}{\frac{1}{\sqrt{3}}} \checkmark$

$$= \frac{1}{\sqrt{2}} \times \frac{\sqrt{3}}{1}$$

$$= \frac{\sqrt{3}}{\sqrt{2}} \checkmark$$



c) LHS = $\frac{\cot \theta \operatorname{cosec} \theta}{1 + \cot^2 \theta}$

$$= \frac{\cot \theta \operatorname{cosec} \theta}{\operatorname{cosec}^2 \theta}$$

$$= \frac{\cot \theta}{\operatorname{cosec} \theta} \checkmark$$

$$= \frac{\cos \theta}{\sin \theta} \div \frac{1}{\sin \theta} \checkmark$$

$$= \frac{\cos \theta}{\sin \theta} \times \frac{\sin \theta}{1} \checkmark$$

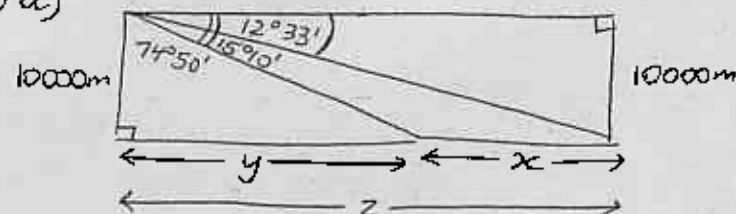
$$= \cos \theta$$

$$= \text{RHS.}$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$1 + \cot^2 \theta = \operatorname{cosec}^2 \theta$$

6 d)



$$\tan 77^\circ 27' = \frac{z}{10000}$$

$$z = 10000 \tan 77^\circ 27' \checkmark$$

$$\tan 74^\circ 50' = \frac{y}{10000}$$

$$y = 10000 \tan 74^\circ 50' \checkmark$$

$$x = z - y$$

$$= 10000 \tan 77^\circ 27' - 10000 \tan 74^\circ 50'$$

$$= 8030.605499 \dots$$

$$= 8031 \text{ m (nearest m).} \checkmark$$

e) $\tan(-150) = -\tan 150$
 $= -\tan(180 - 30)$
 $= -(-\tan 30)$
 $= +\frac{1}{\sqrt{3}}$

Question 5

a) $f(x) = 2x - 1$

i) $f(2) = 2(2) - 1$
 $= 3$ ✓

ii) $0 = 2x - 1$

$1 = 2x$

$x = \frac{1}{2}$ ✓

b) D: $-1 \leq x \leq 1$ ✓
 R: $0 \leq y \leq 1$ ✓

c) $f(x) = \frac{1}{4-x} + \frac{1}{4+x}$

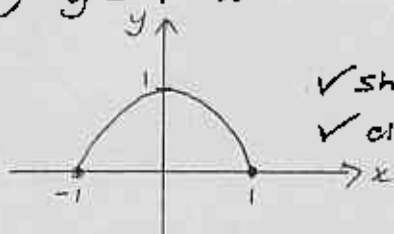
$f(-x) = \frac{1}{4-(-x)} + \frac{1}{4+(-x)}$

$= \frac{1}{4+x} + \frac{1}{4-x}$ ✓

$= f(x)$

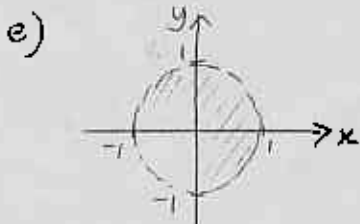
$f(x) = f(-x) \therefore$ even. ✓

d) $y = 1 - x^2$



✓ shape

✓ closed circles at $x = -1, 1$



✓ dotted circle

✓ shading.

(10)

Question 7

a) 345T or N15W ✓

b) $\sin \theta = 0.6018$

acute $\theta = \sin^{-1} 0.6018$

$= 36^\circ 59'$ ✓

$\theta = 36^\circ 59'$ and $180^\circ - 36^\circ 59'$

$= 36^\circ 59'$ and $143^\circ 0'$

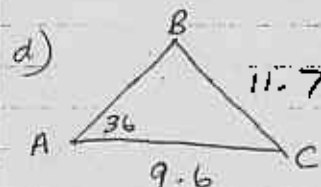
$= 37^\circ$ and 143° (not degree) ✓

c) $\frac{x}{\sin 31} = \frac{14.5}{\sin 85}$ ✓

$x = \frac{14.5 \sin 31}{\sin 85}$

$= 7.496578832...$

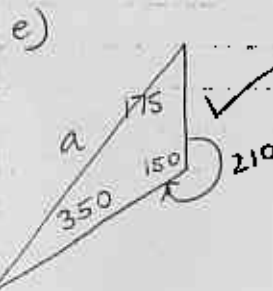
$= 7.5$ (1dp) ✓



$\frac{\sin B}{9.6} = \frac{\sin 36}{11.7}$ ✓

$\sin B = \frac{9.6 \sin 36}{11.7}$

$B = 28^\circ 50' 05.16''$ ✓
 $= 29^\circ$ (not degree)



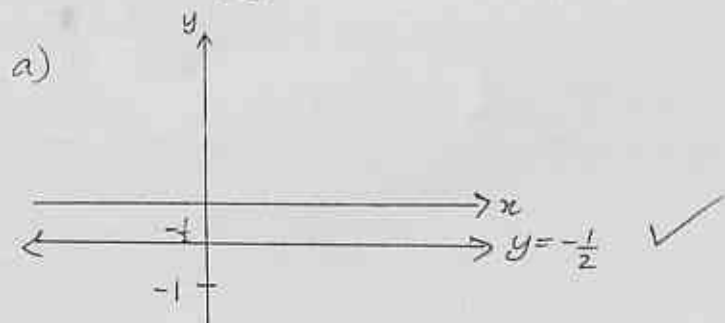
$a^2 = b^2 + c^2 - 2bc \cos A$

$a^2 = 175^2 + 350^2 - 2(175)(350) \cos 150$
 $= 153125 - 122500 \cos 150$
 $= 259213.112$

$a = 509.1297595...$ ✓

$= 509$ nautical miles (not n mile) ✓

Question 8



b) $(-3, 2)$ $(4, 7)$

i) $m = \frac{y_2 - y_1}{x_2 - x_1}$
 $= \frac{7 - 2}{4 - (-3)}$
 $= \frac{5}{7}$ ✓

ii) $d = \sqrt{(y_2 - y_1)^2 + (x_2 - x_1)^2}$
 $= \sqrt{(7 - 2)^2 + (4 - (-3))^2}$
 $= \sqrt{25 + 49}$
 $= \sqrt{74}$ ✓

c) parallel if gradients are same

$y = 3x + 1$ has gradient 3. ✓

$9x - 3y + 2 = 0$

$9x + 2 = 3y$

$3x + \frac{2}{3} = y$ has grad. 3 ✓ parallel.

8d) $(2, -5)$ $(-4, 7)$

midpt: $\left(\frac{2 + (-4)}{2}, \frac{-5 + 7}{2}\right)$

$= \left(-\frac{2}{2}, \frac{2}{2}\right)$

$= (-1, 1)$ ✓

gradient $= \frac{7 - (-5)}{-4 - 2}$

$= \frac{12}{-6}$

$= -2$ ✓

perp bisector will have gradient $\frac{1}{2}$ and go through midpoint. ✓

$y - y_1 = m(x - x_1)$

$y - 1 = \frac{1}{2}(x - (-1))$

$2y - 2 = x + 1$

$x - 2y + 3 = 0$ ✓

e) $3x - 4y + p = 0$ perp dist. to $(2, 5)$ is 2

$2 = \frac{|3(2) - 4(5) + p|}{\sqrt{9 + 16}}$

$2 = \frac{|6 - 20 + p|}{\sqrt{25}}$

$2 = \frac{|-14 + p|}{5}$

$10 = |-14 + p|$

$10 = -14 + p$
 $p = 24$ ✓

or

$10 = -(-14 + p)$

$10 = 14 - p$

$p = 4$ ✓

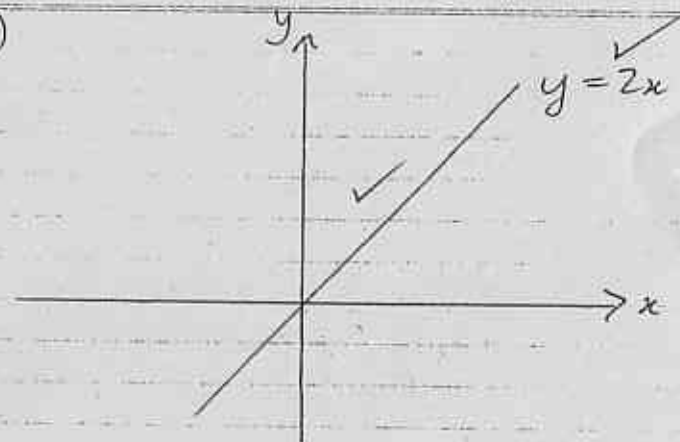
Question 9.

$$\begin{aligned}
 a) \lim_{x \rightarrow -4} \frac{x^4 - 16}{x^2 - 4} \\
 &= \lim_{x \rightarrow -4} \frac{(x^2 - 4)(x^2 + 4)}{(x^2 - 4)} \checkmark \\
 &= \lim_{x \rightarrow -4} x^2 + 4 \\
 &= (-4)^2 + 4 \\
 &= 16 + 4 \\
 &= 20. \checkmark
 \end{aligned}$$

$$\begin{aligned}
 b) \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\
 f(x) = 7x + 2 \\
 f(x+h) = 7(x+h) + 2 \\
 = 7x + 7h + 2
 \end{aligned}$$

$$\begin{aligned}
 \lim_{h \rightarrow 0} \frac{7x + 7h + 2 - (7x + 2)}{h} \checkmark \\
 &= \lim_{h \rightarrow 0} \frac{7h}{h} \\
 &= \lim_{h \rightarrow 0} 7 \\
 &= 7 \checkmark
 \end{aligned}$$

c)



$$\begin{aligned}
 d) f(x) &= 12x^3 \\
 f'(x) &= 36x^2 \checkmark \\
 f'(2) &= 36(2)^2 \\
 &= 144 \checkmark
 \end{aligned}$$

$$\begin{aligned}
 e) y &= 2x^2 + 3x - 1 \\
 \frac{dy}{dx} &= 4x + 3 \checkmark \\
 \text{When } x &= -2 \\
 \frac{dy}{dx} &= 4(-2) + 3 \\
 &= -5 \checkmark
 \end{aligned}$$

Question ⑩

a) $y = 12 + 7x^5$

$y' = 35x^4 \checkmark$

b) $y = (2x-5)^3$

$y' = 3(2x-5)^2 \times 2$
 $= 6(2x-5)^2$

c) $y = x(x-1)$ cuts x axis at $x=0$ $y=0$
 and $x=1$ $y=0$

$y = x^2 - x$

$y' = 2x - 1$

at $x=0$ $y' = 2(0) - 1$
 $= -1$

$x=1$ $y' = 2(1) - 1$
 $= 1$

tangent eqns:

$y - y_1 = m(x - x_1)$

$y - 0 = -1(x - 0)$

$y = -x$

$x + y = 0 \checkmark$

$y - y_1 = m(x - x_1)$

$y - 0 = 1(x - 1)$

$y = x - 1$

$x - y - 1 = 0 \checkmark$

d) $y = x^2$

$\frac{dy}{dx} = 2x$

at $(4, 16)$ grad of tangent is 8
 grad. of normal is $-\frac{1}{8} \checkmark$

e) $y = \frac{x}{x+1}$ $\frac{dy}{dx} = \frac{(x+1) \times 1 - x \times 1}{(x+1)^2} \checkmark$
 $= \frac{1}{(x+1)^2} \checkmark$

⑩